

ARA PATVAKANIAN

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EDUCATION

Visiting Graduate Student in Economics, Harvard University 2025

B.A. in Mathematical Economics & Political Science, University of Pennsylvania 2024

Distinction in Mathematical Economics, Honors Thesis, Summa Cum Laude, Phi Beta Kappa

INTERESTS

Macroeconomics, International Economics, Macro-Finance, Time Series Econometrics

RESEARCH EXPERIENCE

Fellow, Harvard University 2026

Senior Research Associate, Federal Reserve Bank of Boston 2024–2026

- Technical research support for Egon Zakrajšek, Jenny Tang, Slavik Sheremirov, and Danilo Leiva-León
- Provided policy support through data visualization/analysis for 5 presentations to the board of directors of the Boston Fed and 15 pre-FOMC briefings and special topic presentations for Boston Fed’s president
- Data manipulation and model estimation of sign-restricted DFM to decompose PCE inflation into supply- and demand-driven components in WP by [Leiva-Leon et al. \(2025\)](#) and a [Boston Fed policy brief \(2025\)](#)
- Used high-frequency tick data to estimate proxy SVARs for the identification of impulse responses to monetary policy and central bank information shocks in a revision to [Nunes et al. \(2022\)](#)
- Other projects include automating various data processing to model estimation pipelines; computing local projections for policy analysis; identifying tariff pass-through on disaggregated PCE and PPI/IP data; leading trainings for other RAs on VARs and LPs; evaluating the Boston Fed’s internal GDP nowcasting model using pseudo-real-time data; and analyzing the impact of oil price shocks on the U.S. economy.

Research Assistant, University of Pennsylvania 2022

- Used Penn World Tables and Economic Report of the President to visualize inequality and world income distributions in MATLAB for Professors Krueger and Fernández-Villaverde’s textbook manuscript

POLICY PUBLICATIONS

Leila-León, Danilo, Giovanni Olivei, Ara Patvakanian, Jenny Tang, and Egon Zakrajšek. 2026. “[Reassessing the U.S. Economy’s Vulnerability to Oil Shocks](#).” Federal Reserve Bank of Boston Current Policy Perspectives 26-5.

Media Coverage: [U.S. Department of the Treasury](#), [The New York Times](#), [Financial Times](#), [Reuters](#), [Bloomberg](#), [Axios](#)

WORKING PAPERS

“Modeling Macroeconomic Performance through Energy Sourcing Dynamics” 2024

Undergraduate honors thesis advised by Professors Jere Behrman and Francis X. Diebold

- Calibrated structural model of cumulative climate damages to long-run output using historical data
- Tracked lagged effects of damages on future economic performance to inform IAM damage functions

“Predicting Change in Macro-Financial Indicators Using Federal Reserve Statements” 2022

Term paper advised by Professor Karun Adusumilli

- Applied machine learning techniques to evaluate impact of Fed statements on macro-financial indicators
- Used NLP, random forests, and elastic nets to predict movements in inflation expectations and ETFs

COURSEWORK

Harvard University

Graduate Macroeconomic Theory II, Advanced Topics in Macroeconomics (Audited)

University of Pennsylvania

Graduate Microeconomic Theory, Game Theory, Statistical Inference, Topics in Mathematical Logic

Undergraduate Econometric Machine Learning, Bayesian Statistics, Microeconometrics, Real Analysis I & II, Linear Algebra, Differential Equations, Calculus I–III

LEADERSHIP & DISTINCTIONS

Graduation Flagbearer for Mathematical Economics	2024
Penn Economics Undergraduate Advisory Board (Chair)	2022–2024
Penn Undergraduate Economics Society (Co-President)	2020–2024
Penn Armenian Students Association (Co-President)	2020–2024
Resident Advisor, University of Pennsylvania	2022–2024
New Resident Advisor of the Year	2023
Ross Prize for Best Essay in History of Philosophy, Department of Philosophy, U. of Pennsylvania	2024
Feature in Penn Today	2024

MISCELLANEOUS

Technical Python, Stata, MATLAB, R, Shell Scripting, Git, LATEX

Languages Armenian (Native), English (Native), Spanish (Fluent), French (Conversational)

Citizenship USA

REFERENCES

University of Pennsylvania

Francis X. Diebold, Professor

fdiebold@sas.upenn.edu

Dirk Krueger, Professor

dkrueger@sas.upenn.edu

Federal Reserve Bank of Boston

Egon Zakrajšek, Executive Vice President & Director of Research

egon.zakrajsek@bos.frb.org

Jenny Tang, Vice President & Economist

jenny.tang@bos.frb.org